JAY DHABUWALA

LLM Engineer | Agentic AI & RAG Systems | ML Infrastructure

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PROFESSIONAL SUMMARY

Production-focused LLM Engineer with 2+ years at Tata Elxsi building enterprise agentic AI systems, advanced RAG pipelines, and large-scale ML infrastructure. Specialized in multi-agent orchestration (LangChain/LangGraph), hybrid search, and distributed deep-learning training on GPU clusters. Delivered 20% accuracy gains, 30% hallucination reduction, and production predictive systems over 190M+ sample datasets.

CORE TECHNICAL SKILLS

LLM & Agentic AI: LangChain, LangGraph, Multi-agent Orchestration, Tool Calling, Hybrid RAG (BM25 + Vector Search), Structured Output Generation, Adaptive Query Routing, Prompt Engineering

ML & Deep Learning: PyTorch (Distributed Data Parallel), TCN-LSTM / CNN-LSTM time-series models, XGBoost / LightGBM / CatBoost, Captum (Integrated Gradients), EVT/GPD anomaly thresholds, MLflow

Data & Distributed Compute: Databricks, PySpark, Dask, Zarr, 8×H100 GPU training, Unity Catalog, time-series telemetry at 240M+ row scale

Automation & Backend: n8n (Production Mastery certified), FastAPI, Flask-SocketIO, REST APIs, Pydantic, Real-time Streaming, MCP

Cloud & DevOps: AWS (EC2, RDS, ElastiCache, S3, Lambda), Docker, CI/CD Pipelines

Databases & Programming: MySQL, Redis, Pinecone, FAISS, Elasticsearch; Python (Advanced), Node.js, SQL, Pandas, NumPy, Linux/Bash, Git

PROFESSIONAL EXPERIENCE

Engineer — LLM & AI Systems | Tata Elxsi — Mar 2024 – Present

- Architected and deployed multi-agent LLM orchestration systems (LangChain/LangGraph) handling complex document and vehicle-diagnostics workflows.
- Designed hybrid RAG pipelines combining BM25 keyword search with vector similarity search, improving accuracy by 20% and reducing LLM hallucinations by 30%.
- Built production-grade tool-calling systems enabling LLMs to execute database queries, API calls, and file operations with error handling and retry logic.
- Developed a Flask + SocketIO real-time LLM backend serving 50+ concurrent users with streaming responses, message checkpointing, and conversation memory.
- Led ML infrastructure for an EV Battery Management System predictive anomaly-detection program for Stellantis (flagship project detailed below).
- Deployed scalable AI infrastructure on AWS (EC2, RDS, ElastiCache, S3) with automated scaling and health monitoring.

Python Developer | YOU Broadband India Ltd — Sept 2023 – Mar 2024

- Built scalable analytics and automation pipelines processing 100K+ customer records for business intelligence and churn prediction.
- Developed Oracle-based ETL workflows (cxOracle) and CRM-integrated customer re-engagement systems improving retention metrics.
- Created executive dashboards analyzing monthly revenue and variance metrics for C-suite visibility.

KEY TECHNICAL PROJECTS

BMS Predictive Anomaly Detection — EV Battery Management (Stellantis)

PyTorch DDP · Databricks · 8×H100 GPUs · Time-Series Deep Learning · PySpark · MLflow

- Architected a production predictive anomaly-detection system over 190M+ sliding-window samples of multivariate battery telemetry (60-timestep input sequences).
- Trained TCN-LSTM / CNN-LSTM deep-learning models on 8×H100 GPUs via PyTorch Distributed Data Parallel, using Zarr-backed datasets with NVMe and shared-memory caching for high-throughput training.
- Engineered asymmetric anomaly thresholds using Extreme Value Theory (EVT/GPD) per temperature bin, plus distributed gradient-boosting models (XGBoost/LightGBM/CatBoost) via Dask on Databricks GPU clusters.
- Built a DTC confidence-scoring pipeline using Wilson lower-bound lift scoring across 240M+ rows in PySpark, a Captum Integrated Gradients explainability module, and MLflow experiment tracking with Unity Catalog I/O.

Agentic RAG for Enterprise Document Intelligence

LangGraph · Hybrid Search · Multi-agent Validation · Enterprise Scale

- Developed a multi-agent RAG system combining keyword and semantic search with contextual validation and adaptive query routing via LangGraph.
- Implemented agent fallback logic (semantic → keyword → human escalation), delivering reliable, source-attributed answers with confidence scoring across 1000+ enterprise documents.
- **Stack:** LangGraph, Elasticsearch, Vector DB, Claude Sonnet.

Automotive System & Software Requirements Automation (LLM-Based)

LangChain · Multi-agent Validation · ASPICE / Regulatory Compliance

- Engineered an automation pipeline generating complete ASPICE-compliant System and Software Requirements specifications using domain templates, regulatory standards, and constraint-based prompting.
- Reduced requirement-drafting time and improved documentation consistency across engineering teams.
- **Stack:** LangChain, Claude Sonnet & Haiku, multi-agent orchestration.

Team Knowledge Enablement (n8n + RAG)

n8n Automation · RAG Vector Database · S3 Integration

- Designed an end-to-end ingestion pipeline (S3 trigger → n8n workflow → chunking → embedding → vector DB insertion) with file monitoring, format validation, and retry handling.
- Built a RAG chatbot enabling planning, execution, and sales teams to query shared product knowledge, reducing onboarding time and closing cross-team knowledge gaps.
- **Stack:** n8n, Pinecone, FastAPI, S3.

Additional projects: News-to-Script LangGraph pipeline (virality scoring + short-form script generation); PPT automation (LangChain + MCP + python-pptx); LLM chatbot with tool calling, memory & checkpointing (FastAPI / MySQL / Redis); YOLOv8/v11 vision models (95%+ accuracy).

CERTIFICATIONS

n8n Production Mastery · MCP Mastery (Claude, LangChain & Ollama) · Complete MLOps Bootcamp · FastAPI Advanced Web Framework — Udemy

EDUCATION

B.Tech, Computer Science — Visvesvaraya Technological University (VTU) 2019 – 2023 · First Class Distinction (8 CGPA)

ACHIEVEMENTS & RECOGNITION

- Extra Mile Award — Tata Elxsi (2025), for going beyond scope to deliver high-impact LLM solutions.
- Multiple production systems deployed with 99.9% uptime SLA across automotive and enterprise deployments.
- State-Level Swimming Award (Gujarat).

LANGUAGES

English (Fluent) · Hindi (Fluent) · Gujarati (Native)